Quantum Mechanics I — Exercise Sheet 12

Bell's theorem

Variant: Questions only

Conventions, as in Lecture 12: $S = E_{11} + E_{12} + E_{21} - E_{22}$, with E_{xy} the mean of the product of the readings at the settings x and y; the settings of Example 12.5 are $\mathbf{a}_1 = \mathbf{z}$, $\mathbf{a}_2 = \mathbf{x}$, $\mathbf{b}_{1,2} = \frac{1}{\sqrt{2}}(\mathbf{z} \pm \mathbf{x})$. A reading of a spin along a unit vector $\mathbf{a}$ is a reading of $\mathbf{a} \cdot \boldsymbol{\sigma}$.

Exercise 1 (Tsirelson's bound). Let $A_x = \mathbf{a}_x \cdot \boldsymbol{\sigma}$ and $B_y = \mathbf{b}_y \cdot \boldsymbol{\sigma}$ for four unit vectors, and $S_{\text{op}} = A_1 \otimes (B_1 + B_2) + A_2 \otimes (B_1 - B_2)$, so that $S = \operatorname{Tr}(\rho S_{\text{op}})$ for a state ρ of two qubits.

(a) Show that $S_{\text{op}}^2 = 4 \cdot \mathbb{1} - [A_1, A_2] \otimes [B_1, B_2]$.

(b) Show that every eigenvalue of S_{op}^2 is at most 8, and conclude that $|S| \le 2\sqrt{2}$ for every state.

Exercise 2 (three states at the four settings). At the settings of Example 12.5, find S for

(a) the product $|01\rangle$;

(b) the ensemble $\{(\frac{1}{2}, |01\rangle), (\frac{1}{2}, |10\rangle)\}$ of § 10.4;

(c) the density matrix $\rho_p = p\,|\psi^-\rangle\langle\psi^-| + (1-p)\frac{1}{4}\mathbb{1}$, $0 \le p \le 1$ (the singlet with noise). For which p is $|S| > 2$?

Exercise 3 (a local model with perfect anticorrelation). Let the hidden state be a unit vector $\boldsymbol{\lambda}$, uniformly distributed on the sphere, and let the readings be $\alpha = \operatorname{sgn}(\mathbf{a} \cdot \boldsymbol{\lambda})$ and $\beta = -\operatorname{sgn}(\mathbf{b} \cdot \boldsymbol{\lambda})$, a deterministic local model with a continuous hidden state (Remark 12.3).

(a) Show that $E(\mathbf{a}, \mathbf{b}) = -1 + 2\varphi/\pi$, where φ is the angle between $\mathbf{a}$ and $\mathbf{b}$. In particular $E(\mathbf{a}, \mathbf{a}) = -1$.

(b) Find S at the settings of Example 12.5, and compare E with the singlet's $-\cos\varphi$.

Exercise 4 (polariser angles). A photon's polarisation is a qubit, with $|0\rangle = |H\rangle$ (horizontal) and $|1\rangle = |V\rangle$ (vertical). A polariser at the angle ϑ from the horizontal passes $\cos\vartheta\,|H\rangle + \sin\vartheta\,|V\rangle$; record a pass as $+1$ and an absorption as -1.

(a) Show that this reading is a reading of $\cos 2\vartheta\,\sigma_z + \sin 2\vartheta\,\sigma_x$, a spin reading along the Bloch axis at the angle 2ϑ from $\mathbf{z}$ in the xz-plane.

(b) For the pair $\Phi^+ = \frac{1}{\sqrt{2}}(|HH\rangle + |VV\rangle)$, show that $E(\vartheta_A, \vartheta_B) = \cos 2(\vartheta_A - \vartheta_B)$, and find S for the polariser angles $0°, 45°$ on one side and $22.5°, -22.5°$ on the other.

Exercise 5 (Mermin's inequality). Three spins are each read along x or along y, as in § 12.5. Let $E_{xxx}, E_{xyy}, E_{yxy}, E_{yyx}$ be the means of the product of the three readings at these settings, and $M = E_{xxx} - E_{xyy} - E_{yxy} - E_{yyx}$.

(a) Fix a hidden state of a deterministic local model, with the values $x_k, y_k = \pm 1$. Show that $x_1x_2x_3 - x_1y_2y_3 - y_1x_2y_3 - y_1y_2x_3 = \pm 2$. (Write it as $x_1(x_2x_3 - y_2y_3) - y_1(x_2y_3 + y_2x_3)$.)

(b) Show that $|M| \le 2$ in every local model of three spins (12.5).

(c) Find M for ψ_{GHZ}.

(d) Find the reduced state of spins 1 and 2 of ψ_{GHZ}. Can two of the three spins alone violate the CHSH inequality?

Exercise 6 ($\star$ the CHSH game). Two players, who may agree on a strategy in advance but cannot communicate during the game, each receive a random bit $x, y \in \{1, 2\}$, all four pairs with probability $\frac{1}{4}$, and answer $\alpha, \beta = \pm 1$. They win if $\alpha\beta = +1$, except for $x = y = 2$, where they win if $\alpha\beta = -1$.

(a) Show that the winning probability is $\frac{1}{2} + S/8$.

(b) Show that a local model wins with probability at most $\frac{3}{4}$, and that the photons of Exercise 4 win with probability $\frac{1}{2} + \frac{\sqrt{2}}{4} \approx 0.854$.